

Function

In [38]: *# In Python a function is defined using the def keyword:*

```
In [6]: def fun():  
        print('Hello')  
        print('Good Morning Team')
```

```
In [8]: def fun():  
        print('Hello')  
        print('Good Morning Team')  
        greet()
```

Hello
Good Morning Team

```
In [9]: def greet():  
        print('Hello')  
        print('Good Morning Team')  
        greet()  
  
        def greet():  
            print('Hello')  
            print('Good Morning Team')  
            greet()
```

Hello
Good Morning Team
Hello
Good Morning Team

```
In [10]: def greet():  
        print('Hello')  
        print('Good Morning Team')  
        greet()  
  
        print()  
        def greet():  
            print('Hello')  
            print('Good Morning Team')  
            greet()
```

Hello
Good Morning Team

Hello
Good Morning Team

```
In [12]: def greet():  
        print('Hello')  
        print('Good Morning Team')  
        greet()  
        print()
```

```
def greet():
    print('Hello')
    print('Good Morning Team')
greet()
print()

def greet():
    print('Hello')
    print('Good Morning Team')
greet()
print()
```

Hello
Good Morning Team

Hello
Good Morning Team

Hello
Good Morning Team

```
In [14]: def greet():          # declare function without argument
          print('hello')
          print('good morning team')
          greet()
          print('-----')
          greet()
          print('-----')
          greet()              # function calling with out argument
```

hello
good morning team

hello
good morning team

hello
good morning team

```
In [16]: # function without argument
```

```
def greet():
    print('Hello')
    print('Good Morning Team')
greet()
```

Hello
Good Morning Team

```
In [19]: # function with argument
```

```
def add(x,y):
    c=x+y
    print(c)
add(4,5)
```

In [20]: *# function with argument*

```
def add(x,y):
    c=x+y
    return c

add(5,6)
```

Out[20]: 11

In [22]: *# function with argument*

```
def add(x,y):
    c=x+y
    return c

add(5)           # add() missing 1 required positional argument: 'y'
```

```
-----
TypeError                                 Traceback (most recent call last)
Cell In[22], line 7
      4     c=x+y
      5     return c
----> 7 add(5)

TypeError: add() missing 1 required positional argument: 'y'
```

In [23]: *# function with argument*

```
def add(x,y):
    c=x+y
    return c

add(5,6,7)       #add() takes 2 positional arguments but 3 were given
```

```
-----
TypeError                                 Traceback (most recent call last)
Cell In[23], line 7
      4     c=x+y
      5     return c
----> 7 add(5,6,7)

TypeError: add() takes 2 positional arguments but 3 were given
```

In [24]: *# function with argument*

```
def add(x,y,z):
    c=x+y
    return c

add(5,6,7)
```

Out[24]: 11

In [25]: *# function with argument*

```
def add(x,y,z):  
    c=x+y+z+m      # name 'm' is not defined  
    return c  
  
add(5,6,7)
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[25], line 7  
      4     c=x+y+z+m  
      5     return c  
----> 7 add(5,6,7)  
  
Cell In[25], line 4, in add(x, y, z)  
      3 def add(x,y,z):  
----> 4     c=x+y+z+m  
      5     return c  
  
NameError: name 'm' is not defined
```

In [26]: *# function with argument*

```
def add(x,y,z,n):  
    c = x+y+z+m      # name 'm' is not defined  
    return c  
  
add(5,6,7,8)
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[26], line 7  
      4     c = x+y+z+m      # name 'm' is not defined  
      5     return c  
----> 7 add(5,6,7,8)  
  
Cell In[26], line 4, in add(x, y, z, n)  
      3 def add(x,y,z,n):  
----> 4     c = x+y+z+m      # name 'm' is not defined  
      5     return c  
  
NameError: name 'm' is not defined
```

In [27]: *# function with argument*

```
def add(x,y,z,n):  
    c = x+y+z+n  
    return c  
  
add(5,6,7,8)
```

Out[27]: 26

```
In [28]: def greet():
          print('hello')
          print('good morning team')
          greet()

          def add(x,y):
              c = x+y
              return c

          add(5,6)
```

```
hello
good morning team
```

Out[28]: 11

```
In [29]: def greet():
          print('hello')
          print('good morning team')

          def add(x,y):
              c = x+y
              return c

          def sub(x,y):
              d = x-y
              return d

          greet()
          print(add(5,6))
          print(sub(5,6))
```

```
hello
good morning team
11
-1
```

```
In [30]: def add_sub(x,y):
          c= x+y
          d= x-y
          return c, d

          result = add_sub(4,5)
          print(result)
          print(type(result))
```

```
(9, -1)
<class 'tuple'>
```

```
In [31]: def add_sub(x,y):
          c= x+y
          d= x-y
          return c, d

          result, result1 = add_sub(4,5)
```

```
print(result)
print(result1)
print(type(result))
```

```
9
-1
<class 'int'>
```

```
In [32]: def add_sub_mul(x,y):
          c= x+y
          d= x-y
          e= x*y
          return c, d, e

          add, sub, mul = add_sub_mul(4,5)

          add
          sub
          mul
```

Out[32]: 20

UpDate

```
In [34]: def update():
          x = 10
          print(x)
          update()
```

```
10
```

```
In [37]: def update(): # update function take the value from the user
          x = 10
          print(x)

          update(8)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[37], line 5
      2     x = 10
      3     print(x)
----> 5 update(8)

TypeError: update() takes 0 positional arguments but 1 was given
```

```
In [36]: def update(x): #update function take the value from the user
          x = 8
          return x

          update(100)
```

Out[36]: 8

```
In [ ]: def update(x):  
        x = 8  
        return x  
  
a = 15  
update(a)  
print(a)
```